Abstract:

Machine learning (ML)-based anomaly detection is vital for security of networks with heterogeneous traffic. Traditional batch ML models face high maintenance and poor adaptability to rapid anomaly changes, known as concept drift. In contrast, data stream learning integrates incremental learning with concept drift adaptation, enabling seamless updates and quicker recovery of detection performance. This study investigates anomaly detection in streaming IoT traffic as multi-classification, comparing batch Random Forest (RF) with Adaptive Random Forest (ARF), Hoeffding Adaptive Tree (Hoeff) and Gaussian Naive Bayes (NB). Over 80 simulations ARF, Hoeff and NB achieved weighted F1-scores of 0.974, 0.950 and 0.850, outperforming RF (0.549) under drift. Data stream models required less computation than RF, up to half the computation cost for the fastest model. When concept drift adaptation was enabled, recovery of F1-score trace was faster than when only incremental learning was applied, demonstrating the accelerated recovery capabilities of drift adaptation. Finally, we discuss limitations in public IoT datasets, highlighting low traffic heterogeneity, and propose a method to mix existing datasets to simulate realistic data stream scenarios and evaluate model robustness under three types of concept drift: sudden, recurrent and gradual.